

Preverbal infants affirm third-party interventions that protect victims from aggressors

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Protective interventions by a third party on the behalf of others are generally admired, and as such are associated with our notions of morality, justice and heroism^{1–4}. Indeed, stories involving such third-party interventions have pervaded popular culture throughout recorded human history, in myths, books and movies. The current developmental picture is that we begin to engage in this type of intervention by preschool age. For instance, 3-year-old children intervene in harmful interactions to protect victims from bullies⁵, and furthermore, not only punish wrongdoers but also give priority to helping the victim⁶. It remains unknown, however, when we begin to affirm such interventions performed by others. Here we reveal these developmental origins in 6- and 10-month old infants ($N=132$). After watching aggressive interactions involving a third-party agent who either interfered or did not, 6-month-old infants preferred the former. Subsequent experiments confirmed the psychological processes underlying such choices: 6-month-olds regarded the interfering agent to be protecting the victim from the aggressor, but only older infants affirmed such an intervention after considering the intentions of the interfering agent. These findings shed light upon the developmental trajectory of perceiving, understanding and performing protective third-party interventions, suggesting that our admiration for and emphasis upon such acts — so prevalent in thousands of stories across human cultures — is rooted within the preverbal infant's mind.

Imagine the following scenario: a bystander hears a scream and notices a woman being robbed by a man with a gun. The bystander runs to intervene in this aggressive interaction to protect the woman, and the robber flees the scene. To identify this bystander's behaviour as a protective third-party intervention (PTPI) involves (i) understanding that one agent is oppressing another (for instance, that the gunman is robbing the woman); (ii) understanding that there are different consequences of the bystander's action for both the aggressor and victim (the bystander's actions were negative for the robber, whereas they were positive for the woman); and finally, (iii) understanding the goals and/or intentions of the bystander (the bystander appeared to stop the interaction deliberately to help the woman). Therefore, affirming PTPI requires (at least) three interrelated cognitive components: (a) understanding the power relationship between the interacting agents; (b) understanding the value of the bystander's actions for the aggressor and the victim; and (c) understanding the goals and/or intentions of the bystander's action.

Based on current evidence, infants appear to have at least some of these cognitive abilities before the second year of life, suggesting that they may have the necessary mechanisms for affirming PTPI. For example, infants can recognize the goals of an agent's action^{7–9}. Moreover, infants can discriminate between positive and negative interactions involving two agents¹⁰ and also evaluate an agent's actions in terms of their helping or hindering the goals of the other agent involved in such interactions¹¹, even in a second-order manner (negative behaviours directed to a bad agent are perceived by infants as positive¹²). Furthermore, not only do infants infer dominance hierarchies (the strong and the weak) from both body size¹³ and social interactions¹⁴, but they can also discriminate the victim from the aggressor in aggressive interactions¹⁵.

To explore the affirmation of PTPI in relation to its possible early development and underlying mechanisms, we designed (and subsequently modified) a new paradigm involving both 6- and 10-month-old infants, who watched an animated scenario involving an interaction between (i) an aggressor and (ii) a victim along with (iii) a bystander that either interfered to offer protection or did not interfere (see Fig. 1 for details). Consistent with our emphasis on protection for others, in this paradigm third-party intervention is considered to include some risk because infants regard such aggressive interaction as negative¹⁶ and avoid the aggressor¹⁵. We first addressed whether 6-month-olds can in fact affirm PTPI by using our new paradigm in conjunction with a forced choice task to see whether they preferred the interfering rather than the non-interfering agent depicted in the prior animation.

In Experiment 1, 20 6-month-old infants were alternatively familiarized with two types of animation during which a coloured cubic geometric agent with eyes watched an aggressive interaction between two spherical geometric agents with eyes; this concluded when the cube agent either interfered in the interaction by inserting itself between the two spherical agents (interfering event), or did not interfere (non-interfering event) (Fig. 1a, Supplementary Videos 1 and 2). In a subsequent forced choice task, infants were encouraged to choose between two real cubes with eyes which also corresponded to the differing colours used in the two animations (for example, the interfering cube was green, whereas the non-interfering cube was orange). If infants affirm PTPI then they should choose the interfering rather than non-interfering agent.

In support of this prediction, we found that infants chose the interfering agent significantly more than the non-interfering agent (17 of 20 infants, binomial test, two-tailed, $P < 0.003$; Fig. 2a). These findings are not attributable to differences involving attention

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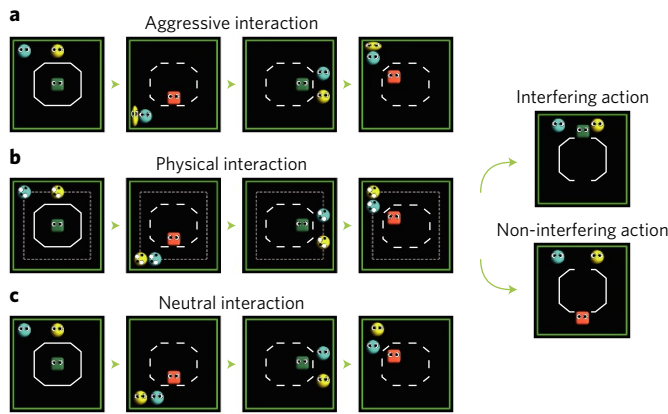


Figure 1 | Selected frames from the animated scenarios in Experiments 1 to 3. a, Aggressive interaction in Experiment 1 (Supplementary Videos 1 and 2). **b,** Physical interaction in Experiment 2 (Supplementary Video 3). **c,** Neutral interaction in Experiment 3 (Supplementary Video 4).

(looking more at scenarios involving an interfering rather than non-interfering agent) as we found no significant differences for looking time between animations either for the peripheral area of interest, AOI ($t(19)=0.554$, $P=0.593$), or central AOI ($t(19)=-0.420$, $P=0.679$); see Supplementary Fig. 1, and Supplementary Table 1. Our findings, therefore, indicate that 6-month-olds do have the ability to affirm PTPI, as indicated by their preference for choosing the interfering agents observed within an aggressive interaction. To rule out alternative lean explanations for our results, we conducted further experiments using variations of our basic paradigm.

It has been proposed that infants may consider collisions involving geometric shapes merely as a negative physical event rather than being a negative social interaction¹⁶. Thus, it is possible that infants in our study regarded the interaction between the spherical figures in mere physical rather than socially aggressive (animate) terms, and as a result preferred the agent that stopped the negative physical event rather than the aggressive interaction *per se*. In Experiment 2, we tested this possibility by recruiting additional 6-month-old infants ($N=20$) who were also familiarized with animations identical to those in Experiment 1 with the following exceptions: we eliminated the perceivable ‘social animacy or agency’ of both the interacting spheres by making them appear as if they (i) had no eyes (rather than having eyes), (ii) were non-self-propelled (rather than self-propelled), and (iii) involved no distortion on contact (rather than showing distortion on contact) (Fig. 1b, Supplementary Video 3). If infant’s preference for PTPI is driven by their mere preference for the agent that stopped the unpleasant physical event, we would expect infants to show a preference for the interfering rather than non-interfering agent. However, if infant’s protective preferences are driven by their perception of an aggressive social interaction (as in Experiment 1), we would expect no preference for either interfering or non-interfering agents.

Consistent with this latter prediction, there was no significant difference in infants’ choosing between the interfering agents and non-interfering agents (10 of 20 infants, binomial test, two-tailed, $P=1.000$, Fig. 2a). These findings are not attributable to differences involving attention, as we found no significant differences for looking times between animations either for the peripheral AOI ($t(19)=-1.268$, $P=0.220$) or central AOI ($t(19)=1.130$, $P=0.273$) (see Supplementary Table 1). This result significantly differed from the forced-choice behaviour reported in Experiment 1 (Fisher’s exact test, two-tailed, $P<0.041$, Fig. 2a). Subsequently, we have excluded the possibility that infants’ choices in Experiment 1 were driven by their mere preference for the agents’ simply stopping a negative event rather than for animate socially interacting agents.

It remains possible, however, that an infant’s preference for the agent who did PTPI in Experiment 1 was simply due to the fact that the interfering agent approached and interacted with the other agents, who themselves were merely interacting socially regardless of whether it was negative. Therefore, perhaps the infants’ preference for the interfering agent was due to their regarding these agents as being merely social. To exclude this possibility, in Experiment 3 we again tested another group of infants ($N=20$) with a modified version of the animation used in Experiment 1, but this time we made the interaction between the spherical agents appear less negatively or positively ‘social’: the two spheres simply moved together in a chase–chaser interaction with some degree of contact (rather than involving more perceptually salient collisions) (Fig. 1c, Supplementary Video 4). If infants’ preferences for PTPI were driven by their regarding the interfering agent as acutely social, then we would expect them to prefer the interfering rather than non-interfering agent. However, if infants’ protective preferences were driven by their perception of the interfering agent acting to stop a negative social interaction, then we would predict no preference in their choices between interfering and non-interfering agents.

In support of this latter prediction, no significant preference for the interfering agent was found in infants’ choice behaviour (8 of 20 infants, binomial test, two-tailed, $P=0.503$; Fig. 2a). These findings are not attributable to differences involving attention, as we found no significant differences for looking times between animations either for the peripheral AOI ($t(19)=-0.404$, $P=0.690$) or central AOI ($t(19)=-0.782$, $P=0.444$) (see Supplementary Table 1). This outcome is significantly different from the choice behaviour of Experiment 1 (Fisher’s exact test, two-tailed, $P<0.008$, Fig. 2a), and therefore we have excluded the lean explanation suggesting a preference for acutely social interfering agents. Taken together, our results from these three experiments suggest that 6-month-olds can affirm PTPI actions (Experiment 1), and we have ruled out factors involving preferences for merely stopping a negative event (Experiment 2) and for acute sociality (Experiment 3), respectively.

Although PTPI affirmation seems to emerge in early infancy, it remains unclear whether infants identified the action in relation to the three psychological characteristics proposed above: understanding the power relationship between the interacting agents; understanding the value of the third-party bystander’s actions for the aggressor and the victim; and understanding the goals and/or intentions of the bystander’s action. Let us first consider the power relationship. Did infants perceive the interfering action as protecting the victim from the aggressor; stated differently, that the victim was the victim and the aggressor was the aggressor? Results from Experiments 1–3 suggest that this was the case. Recall that in Experiment 2, we addressed the possibility that infants have a preference for anything that intervenes in a situation involving physical contact. Our results excluded this possibility and in doing so reaffirmed the likelihood that infants perceived the interaction in Experiment 1 to involve aggressive social interaction. Moreover, Experiment 3 revealed that infant’s original preferences were driven by perceiving a negative social interaction rather than any social interaction *per se*. Hence, we can infer that infants understood the interaction in Experiment 1 to involve an aggressor and a victim. This leads us to consider whether infants understood the value of the third-party bystander’s action for both the aggressor and victim. If so, then infants should be able to infer that the interfering action has different consequences (and hence values) for each spherical agent: for the victim, the consequence of the interfering action was protection from harm and hence positive; for the aggressor, the consequence of the interfering action was stopping the attempt to harm and hence negative.

Unlike prior infant studies investigating the values of an action^{11,12}, Experiments 1–3 did not include an explicit outcome of the interfering action (for example, taking the victim away) and hence do

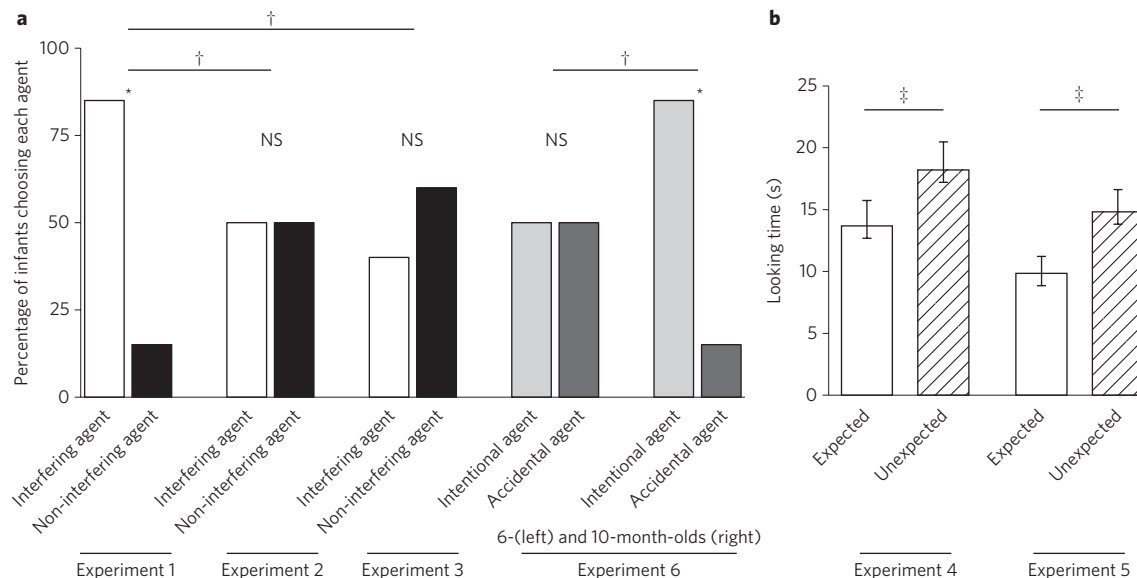


Figure 2 | Results from Experiments 1 to 6. a, Results of forced-choice tasks across experiments (Experiments 1 to 3, and 6). Asterisk: binomial test, two-tailed $P < 0.05$; dagger: Fisher's exact test, two-tailed $P < 0.05$; NS, non-significant. **b**, Mean looking time and standard errors for the expected and the unexpected events in Experiments 4 and 5. Double dagger: paired t -test, two-tailed $P < 0.05$.

not allow us to make any conclusions about the infants' evaluations. Such inferences, however, may be reflected by infants' looking patterns for when the bystander interacts with both spherical agents after the interference has occurred. More specifically, if the bystander subsequently interacts with each agent in a congruent manner to what the infant expects based on what happened during the interfering scenario (for example, the bystander interacts with the aggressor in an aggressive manner), the infants should show little interest as their expectations have not been violated; conversely, if the bystander subsequently interacts with each agent in an incongruent manner to what the infant expects (for example, the bystander interacts with the aggressor in a helpful manner), the infants should show much more interest.

In Experiments 4 and 5, using our original animation paradigm in conjunction with expectation-violation methods, we explored for the presence of these two underlying evaluations in relation to infants' preference for interfering agents. More specifically, following the original animation scenarios, infants were then presented with another one out of two possible scenarios. In Experiment 4, the interfering agent attacked both the aggressor (a negative action towards the aggressor) and the victim (a negative action towards the victim; Fig. 3, Supplementary Videos 5 and 6). In Experiment 5, the interfering agent helped both the aggressor (a positive action towards the aggressor) and the victim (a positive action towards the victim; Fig. 3, Supplementary Videos 7 and 8). Testing with adults confirmed that these animations were being perceived as intended (that is, positive actions; see Supplementary Discussion for details).

In support with our above predictions, results across both experiments found that infants looked significantly longer at the unexpected events than at the expected events (Experiment 4; $t(15) = -3.285$, $P = 0.005$, Experiment 5; $t(15) = -3.097$, $P = 0.007$, Fig. 2b) (see Supplementary Table 2). In both experiments, the aggressor was always presented on the left and the victim always on the right; we did this to reduce the task load for the infant. However, we can rule this potential confound out as we observed differential looking patterns across experiments (looking at the right side for longer when the victim was being attacked by the bystander, and longer at the left side when the aggressor was being helped by the bystander). Consistent with the notion that affirming PTPI requires understanding the value of the bystander's actions, these results therefore

suggest that infants do in fact form two different evaluations for the interfering action according to the different roles of the acted agents (the aggressor and victim). This therefore also reaffirms that infants do understand the power relationship between these agents.

Finally, there is the issue of goal and/or action understanding; after all, we tend to admire or emulate people whose protective actions are deliberate rather than accidental. Therefore, PTPI should only be affirmed as such when they involve deliberate rather than accidental actions. Are infants capable of such perspective taking? Although the difference between understanding goals and intentions may seem trivial, the latter reflects situations in which the final objective of an agent is not clear until after it is completed, whilst the former makes it clear what the agent's objective is prior to being completed^{17,18}. Prior research has established that 6-month-olds understand goal-directed actions⁷⁻⁹, but the emergence of understanding intentions seems to first emerge by age 8 months¹⁸, although we note that there have been very few studies investigating the latter in young infants. We therefore chose to investigate intention understanding within the context of affirmation for PTPI in both 6-month-olds ($N = 20$, who based on Experiment 1 show affirmation for such acts) and 10-month-olds ($N = 20$, who are slightly older than those found to show intention understanding).

To do so, we modified our original animation plus forced-choice paradigm. In the 'active intervention' familiarization trials, the cube agent in a barrier followed the two spherical agents moving around the outer side of the barrier, and although it appeared that it was trying to somehow interact with the spheres (but could not because of the barrier), the actual objective of this action was not clear until the very end when it inserted itself between the spheres rather than doing something else (for example, attacking one of the spheres). In the 'non-active intervention' trials, the cube did not follow or interact with the spheres at all, until right at the end when interference did occur. Therefore, although the outcome was the same in both conditions, the supposed objectives were different (interacting versus non-interacting; Fig. 4, Supplementary Videos 9 and 10). We confirmed that differences in perceived intentionality existed by testing adults (see Supplementary Discussion for details). If infants can take into account the intention of the bystander's action, we expected that they would prefer choosing the active interventional agent rather than the non-active interventional agent.

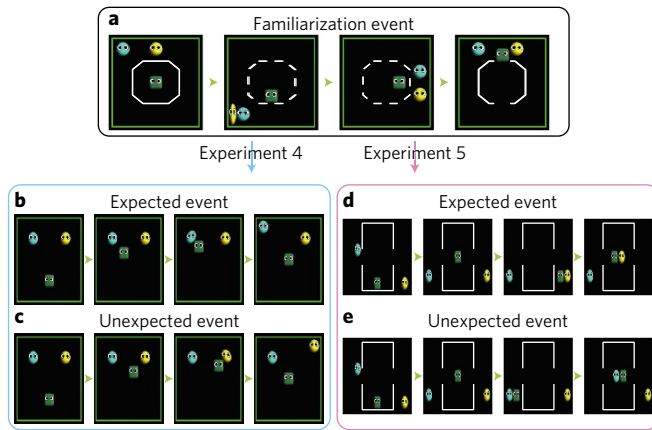


Figure 3 | Selected frames from the animated scenarios in Experiments 4 and 5. **a**, Familiarization event (Supplementary Video 1). **b**, Expected event of Experiment 4 (Supplementary Video 5). **c**, Unexpected event of Experiment 4 (Supplementary Video 6). **d**, Expected event of Experiment 5 (Supplementary Video 7). **e**, Unexpected event of Experiment 5 (Supplementary Video 8).

In partial support of our prediction, we found that whereas 6-month-olds did not show any differences in their choices between active-interventional or non-active-interventional agents (10 of 20 infants, binomial test, two-tailed, $P = 1.000$, Fig. 2a), 10-month-olds robustly chose the active interventional rather than the non-active-interventional agent (10-month-olds; 17 of 20 infants, binomial test, two-tailed, $P < 0.003$). These outcomes reflect a significant difference for choice behaviour between the two ages (Fisher's exact test, two-tailed, $P < 0.041$, Fig. 2a). Such differences are not attributable to attention, as we found no main effects involving looking time for either 'intervention' (active versus non-active), 'age' (6- versus 10-month-olds) or their interaction (peripheral AOI, all F values (1, 38) < 1.119 , all P values > 0.296 ; central AOI, all F values (1, 38) < 0.963 , all P values > 0.333 ; see Supplementary Table 3). Further, given the results of Experiment 3, the choices of 10-month-olds cannot be explained by a preference for mere sociality. Therefore, these results show that 10-month-olds but not 6-month-olds could evaluate the protective action based on the agent's apparent intentions, even if the outcomes of the interfering actions were the same. This finding is consistent with a single prior infant study¹⁸ indicating that only during the second half of their first year do infants begin to evaluate the intention of actions regardless of the outcomes. In summary, Experiments 4–6 indicate that young infants do evaluate PTPI in terms of power relationships and the values of the action itself, although it appears that only 10-month-olds can do so in relation to understanding intentions rather than simple goal-directed actions.

Our findings show that the affirmation of PTPI is already evident in preverbal infants: those as young as 6 months preferred the interfering agents to non-interfering agents after an aggressive interaction. Such a preference, which could not be attributed to their possible preferences for merely stopping a negative event (Experiment 2) or acute sociality (Experiment 3), also appears to involve correct understandings for both power relationships (Experiments 1–3) and the value of the interfering action (Experiments 4 and 5). However, although 6-month-old infants appeared to show affirmation for PTPI, unlike 10-month-olds their understanding is relatively rudimentary owing to their inability to differentiate correctly between active and non-active interventional actions (Experiment 6). Nonetheless, our findings suggest that affirming PTPI begins much earlier than perhaps expected. Indeed, although preschool children interfere in harming interactions to protect the victim from bullies⁵,

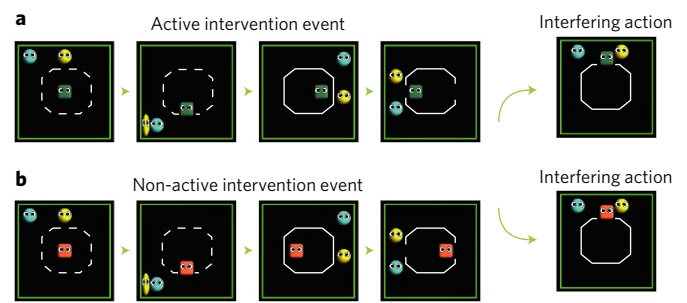


Figure 4 | Selected frames from the animated scenarios in Experiment 6. **a**, Active intervention event (Supplementary Video 9). **b**, Non-active intervention event (Supplementary Video 10).

and not only punish wrongdoers but also give priority to helping the victim⁶, it seems that such abilities have long been preceded by what appears to be their foundational ability to affirm such actions. Detailing how we go from affirming to performing PTPI is a challenge for future researchers. Furthermore, determining how and when children incorporate their abilities for PTPI into their developing notions of morality, justice and heroism^{1–4} will ultimately be required.

Our findings add to the growing evidence concerning preverbal infants' basic ability for evaluating the actions of others. Prior studies have typically asked whether preverbal infants evaluated others' actions within an interaction between two agents (for example, one agent helps the other to climb a hill by pushing^{11,12}); thus, the action to be evaluated has one explicit value (pushing up the hill is always seen as positive; there is no negative value for pushing). The evaluation process in the current study is derived from a more complex triangular relationship in which the same action to be evaluated potentially has two different values (positive for the victim but negative for the aggressor). Our findings nevertheless indicate that 6-month-old infants can attribute such different values involving the interfering action. This means that young infants do have the ability to recognize the relative values of one action (for example that the same action is good for one but bad for another) rather than just understanding the absolute values for one action (one action has one value for all others); in other words, young infants possess a more complex evaluation system of others' actions beyond that required for simple, unambiguous situations. Moreover, unlike studies that have emphasized simple helping behaviour or reciprocation^{11,12,18}, our series of experiments involved interactions emphasizing risk and protection, which requires more complex cognitive abilities as mentioned above.

Furthermore, how do infants attribute different (positive or negative) values to the same interfering action as seen in Experiments 4 and 5? One possibility is that infants may evaluate such bystander's actions when motivated by both restorative (sympathetic) and retributive (punishment) justice^{2,3}, both of which are evident in prior infant research (for instance, infants show a sympathetic attitude to others in distress¹⁵ and also a preference for agents who punish wrongdoers¹²). Perhaps infants' positive and negative evaluations of the interfering action are respectively driven by restorative and retributive motivations, although future research is required to disentangle whether and how these underlying motivations are in fact related to PTPI affirmation.

Close inspection of our findings involving infant preference for intervening agents in an aggressive interaction raises an important related issue: rather than assuming that this agent's intervention would lead to a negative interaction with the victim, it appears that infants assume that it will be a positive interaction (recall that the intervention simply involved the agent placing itself between the aggressor and the victim; there was no interaction between the

agent and either of these actors). If so, why do infants seem to have this as their default assumption? Perhaps motivations associated with restorative and/or retributive justice are of relevance here too, but this remains to be explored.

Finally, our developmental insights about the necessary and/or sufficient mechanisms associated with PTPI affirmation can be complemented with a phylogenetic perspective. Possible phenomena resembling the performance of protective third-party actions have been described as ‘policing’ (that is, an impartial intervention in conflicts) in non-human primates. Such behaviour involves high-ranking individuals^{19,20} whose absence results in increased conflicts among the remaining group members²¹. Investigating such behaviours and their perception by conspecifics would shed a light not only on whether PTPI affirmation is uniquely human, but also on the possible selection pressures and adaptive functions associated with such a prosocial ability. Indeed, we note that there are several proposals involving our potentially being hardwired for early pro-sociality in conjunction with environmental influences²² (for example, not only do 1–2-year-old infants show different neural signatures when watching prosocial and antisocial acts, but these differences are related to their parents’ sensitivity to injustice for others²³). In any event, our admiration for PTPI — and the thousands of stories that have resulted — seems rooted in the infant’s mind.

Methods

Experiment 1. Participants. Participants were 20 healthy full-term 6-month-old Japanese infants (males = 10; mean (*M*) age = 6 months 12 days, range 6.00–6.28 months). Prior infant morality studies involve similar numbers^{11,15,16,18}. Two additional infants were tested but excluded because of distress or fussiness (1) or experimental error (1). Parents provided written informed consent prior to the experiment and were financially compensated for participation. The experiment was approved by the Ethics Committee of Kyoto University, Kokoro-Unit (no. 26-P-10).

Stimuli. Two familiarization animation scenarios were created using Poser 6.0 (e-frontier Inc.). In the familiarization scenario, each geometric cube agent (green or orange) with eyes, which was within a white bounded enclosure, watched an aggressor–victim interaction outside this enclosure in which one geometric spherical figure with eyes (aqua coloured) hit and crashed into another similar-looking (but yellow) figure. After this aggressive interaction, the cube agents (green and orange) either interfered in the interaction at the top of the display (interfering event) or did not interfere at the bottom of the display (non-interfering event) (Fig. 1a, Supplementary Videos 1 and 2). The two types of familiarization events (41 seconds in duration) were presented four times in alternate order (that is, the roles of the different coloured cubes and the presentation orders of the two familiarization animations were counterbalanced between participants). Following the familiarization phase was a forced-choice task in which infants were presented with two real cube objects corresponding to the colours and features seen in the animation (each cube was 6.5 × 6.5 × 6.5 cm). The presentation position of these two objects (left or right) was counterbalanced between participants.

Materials and procedure. A Tobii TX300 (Tobii Technology) near-infrared eye-tracking system recorded infants’ gaze movements at 60 Hz. An integrated 23-inch computer display (1,920 × 1,080 pixels) presented the animated stimuli in a darkened room by means of a computer (Precision M6600, Dell Japan) with Tobii Studio software (Tobii Technology). Infants sat on their mothers’ laps about 60 cm from the monitor. A five-point calibration was administered. Parents did not talk or interact with the infants during the experiment.

Randomly allocated infants were presented with eight familiarization trials. Attractive animated clips with sound (for example a moving blue bear) were inserted between trials to maintain attention towards the monitor. Following familiarization trials, parents rotated their chairs clockwise 90° from the monitor to position their infants sitting straight ahead on the front of their laps, and then closed their eyes. In the forced choice task, an experimenter blind to the actions of the two geometric figures presented two real cubes (about 18 cm apart) on a black board (42 cm × 30 cm) about one metre in front of the infants, encouraging them to choose between these two objects. After confirming that infants looked at both objects, the experimenter moved the objects closer to the infants. This was recorded by a video camera.

Data analysis. Parametric tests were used when assumptions were met. Gaze data were analysed using Tobii’s standard statistics package (Tobii Technology). Gaze acquisition data below 50% indicated fussiness and were excluded.

Familiarization trials involved two AOI: one covering the interacting geometric spherical agents outside the white enclosure (inner boundary) but inside the green enclosure (outer boundary) (that is, ‘peripheral AOI’); one covering the interfering geometric cube agents inside the white enclosure (‘central AOI’) (see Supplementary Fig. 1). The mean looking times at each AOI during familiarization trials are in Supplementary Table 1.

In the forced choice measure, each infant’s preference towards the real cubes was measured by recording the one that was first contacted using a visually guided reach¹⁸. Inclusion required that infants contact a single object, not two objects. All choices were coded while blind to the condition. A second coder, blind to the purpose of the experiment, rated half of these trials selected at random. Coders showed perfect agreement (Cohen’s $\kappa = 1.00$).

Experiment 2. Participants. Participants were an additional healthy 20 full-term 6-month-old infants (males = 10; *M* age = 6 months 15 days, range 6.03–6.28 months). Four additional infants were tested but excluded owing to distress or fussiness (3) or failure to meet the reaching criteria (1). All other details are as in Experiment 1.

Stimuli. The animations in Experiment 2 were identical to those of Experiment 1, with the following exceptions: we eliminated the perceivable ‘social animacy or agency’ of both the interacting cubes by making them appear as if they (i) had no eyes, (ii) were non-self-propelled²⁴, and (iii) involved no distortion on contact (Fig. 1b, Supplementary Video 3).

Materials and procedure. The materials and procedure were identical to those of Experiment 1.

Data analysis. Gaze data were analysed using Tobii’s standard statistics package. The mean looking times for each AOI during familiarization trials are in Supplementary Table 1. The forced choice criteria and analyses were the same as for Experiment 1. Coders showed perfect agreement (Cohen’s $\kappa = 1.00$).

Experiment 3. Participants. Participants were an additional 20 healthy full-term 6-month-old infants (males = 10; *M* age = 6 months 16 days, range 6.02–6.28 months). Three additional infants were tested but excluded owing to distress or fussiness (1), parental interference (1) or failure to meet the reaching criteria (1). All other details are as in Experiment 1.

Stimuli. The animations in Experiment 3 were identical to those of Experiment 1, but this time we made the interaction between the spherical agents appear less negatively or positively ‘social’: the two spheres simply moved together in a chase–chaser interaction with some degree of contact (rather than involving more perceptually salient collisions) (Fig. 1c, Supplementary Video 4).

Materials and procedure. The materials and procedure were identical to those of Experiment 1.

Data analysis. Gaze data were analysed using Tobii’s standard statistics package. The mean looking times at each AOI during the familiarization trials are in Supplementary Table 1. The forced choice criteria and analyses were the same as for Experiment 1. Coders showed perfect agreement (Cohen’s $\kappa = 1.00$).

Experiment 4. Participants. Participants were an additional 16 healthy full-term 6-month-olds (males = 12; *M* age = 6 months 12 days, range 6.03–6.27 months). Two additional infants were tested but excluded owing to their failure to meet inclusion criteria (1) and fussiness (1). All other details are as in Experiment 1.

Stimuli. The familiarization scenarios in Experiments 1 and 4 were identical. Following this was another animation in which the interfering agent (i) separately attacked either the aggressor or the victim three times (that is, a collision into the aggressor/victim as reported in previous studies²⁵) (Fig. 3, Supplementary Videos 5 and 6), before (ii) returning to the central position below and between these two spherical agents. At this point, the animation froze for a maximum of 45 s. The positions of the aggressor (left) and the victim (right) were fixed so as not to confuse infants. Total duration time for the second animation was 59.69 s. The colours of the interfering agents and the order in which they were first attacked were counterbalanced between participants.

Materials and procedure. The materials and procedure were identical to those of Experiment 1 with the following exceptions. First, during familiarization infants were presented four times (rather than eight times) solely with the ‘interfering event’ of Experiment 1. Second, in subsequent test events, infants were alternatively presented four times with two additional animations (‘expected event’ and ‘unexpected event’). The presentation order of these two test events was counterbalanced between participants. Third, a web camera (C920R, Logitech) was additionally attached to the centre of the upper frame of the display monitor so that we could observe participants’ looking behaviour via a computer (CF-SX1, Panasonic Corporation). During gaze measurement, an experimenter who was

blind to the experimental condition monitored the infant's gaze to the main display in the test events and measured it with a customized program. A given test trial was considered to be completed when the infant (i) looked away for two consecutive seconds during the frozen segment of the second video, after (ii) having first looked for at least two consecutive seconds from the start of the colliding interaction (which was the length of time it took for the interfering agent to direct itself towards and initially collide with one of the two other agents). Fourth, there was no forced-choice task.

Data analysis. Gaze data were analysed using Tobii's standard statistics package. The mean looking times in each of the familiarization and test trials are in Supplementary Table 2. Following procedures used in previous studies^{26,27}, we based our analyses on data from the infant's first completed pair of trials (that is, those trials in which the minimum looking criteria described above were first met). The looking times in test trials were measured from the starting point of the videos when the interfering agents moved towards the aggressor or the victim, and finished on the infant looking away for 2 seconds after the cube returned to its original position (the frozen segment).

Experiment 5. Participants. Participants were an additional 16 healthy full-term 6-month-old infants (males = 4; *M* age = 6 months 14 days, range 6.00–6.27 months). Five additional infants were tested but excluded owing to failure to meet inclusion criterion (1), experimental error (1), calibration error (1) and fussiness (2). All other details are as in Experiment 1.

Stimuli. The familiarization videos in Experiments 1 and 5 were identical. Following this was another animation in which the interfering agent separately helped either the aggressor or the victim (by joining, caressing and guiding the aggressor/victim into the middle of the enclosed barrier¹⁰) (Fig. 3, Supplementary Videos 7 and 8). At this point the video froze for a maximum of 45 s. The total duration time for this second animation was 75.47 s. The colours of the interfering agents, along with which agents first started to enter the white enclosure and the order in which assistance occurred, were counterbalanced between participants. We confirmed the validity of these animations using adults (see Supplementary Discussion).

Materials and procedure. The materials and procedure were identical to those of Experiment 4 with one exception. A given test trial was considered to be completed when the infant (i) looked away for two consecutive seconds during the frozen segment, after (ii) having first looked for at least six consecutive seconds from the start of the trial (which is the length of time it took for both the aggressor and the victim to unsuccessfully attempt to enter the white enclosure) and (iii) having also first looked for at least two consecutive seconds after the interfering agent actually contacted one of the spherical agents (which was the length of time it took for the interfering agent to caress one of the two other agents).

Data analysis. The data analysis were identical to those of Experiment 4 except that the looking times in test trials were measured after 15.8 s had passed from the starting point of videos (from when the interfering agents moved to the aggressor or the victim). The mean looking times in the familiarization and test events are in Supplementary Table 2.

Experiment 6. Participants. Participants (40 in total) were an additional 20 healthy full-term 6-month-old infants (males = 10; *M* age = 6 months 14 days, range 6.04–6.27 months) and 10-month-old infants (males = 10; *M* age = 10 months 15 days, range 10.01–10.29 months). Two additional 6-month-olds were tested but excluded owing to distress or fussiness (1) or calibration error (1). Three additional 10-month-olds were tested but excluded owing to distress or fussiness (2) or failure to meet our reaching criteria (1). All other details are as in Experiment 1.

Stimuli. The animated segments in Experiments 1 and 6 were identical, with the following exceptions. In the 'active intervention' familiarization trials, the cube agent followed the two spherical agents around the screen, and although it appeared that it was trying to somehow interact with the spheres (but could not because of the barrier), the actual objective of this interaction was not clear until the very end when it inserted itself between the spheres rather than doing something else (for instance, attacking one of the spheres). In the 'non-active intervention' trials, the cube did not follow or interact with the spheres at all, until right at the end when interference did occur. Therefore, although the outcome was the same in both conditions, the supposed objectives were different (active intervention versus non-active intervention; Fig. 4, Supplementary Videos 9 and 10). We confirmed the validity of these animations using adults (see Supplementary Discussion).

Materials and procedure. The materials and procedure were identical to those of Experiment 1.

Data analysis. Gaze data were analysed using Tobii's standard statistics package. The mean looking times for each AOI during familiarization trials are in

Supplementary Table 3. The forced choice criteria and analyses were identical to Experiment 1. Coders showed perfect agreement (Cohen's $\kappa = 1.00$).

Data availability. The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Y.K., Y.I. and G.M. designed the study, supervised by K.H and M.M.-Y. Y.K. performed the experiments. Y.K. analysed the data. Y.K. and D.B. drafted the paper, and all authors discussed the results and commented on the final manuscript.

Additional information

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Competing interests

The authors declare no competing interests.